

EN525.733-8VL2+8VL3: Deep Learning for Computer Vision

Syllabus

Spring 2025 – 3 Credit Hours

Class Meetings Zoom

Instructor: Prof. Nasser M. Nasrabadi

Classroom: Virtual Lectures, Tuesdays and Thursdays: 7:15-10pm

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Credit Hours: 3 credit hours

Pre-requisites: CPE 520 or equivalent, Graduate standing

Textbook & Course Materials:

- “Deep Learning” by Ian Goodfellow, Yoshua Bengio, and Aaron Courville, MIT Press, Nov 18, 2016, ISBN:0262035618, 9780262035613. **(Optional)**
- Class Notes will be provided
- Journal & Conference papers will be assigned for HWs and class Projects

Required and optional instructional materials: None

Course Description:

A study of deep learning architectures and their underlying theory. Review of neural network architectures; introduction to advanced deep neural network architectures that use many layers and large image databases; application of deep learning to dimensionality reduction, latent feature extraction, and manifold representation; coupled deep neural networks for cross-modality object verification; use of multiple neural networks for data fusion; applications of deep learning in biometrics, computer vision, and data mining.

Course Objectives:

The purpose of this graduate course is to provide the fundamental set of techniques and advanced architectures that constitute deep learning as of today, while providing a thorough grounding in the methodologies and mathematical foundations.

Expected Learning Outcomes:

- Understand the modern CNN-based architectures (i.e., LeNet, AlexNet, VGGNet, GooGleNet, ResNet, DenseNet, HighwayNet, and FractalNet).

- Describe the relative merits of various deep learning architectures (i.e., CNN, GAN, Cycle GAN, and LSTM).
- Explain auto-encoders for feature extraction, dimensionality reduction and manifold representation.
- Use of regularization for network sparsity.
- To be able to use advanced deep learning frameworks for object recognition, pedestrian detection, face, iris, and fingerprint identification.

Class Format: The course consists of lectures with discussions, reading assignments (technical papers), one mid-term, one extensive final project, and 8 homework assignments using dedicated neural networks software such as Pytorch, Tensor Flow, Caffe, or Matlab. Each student will carry out one independent mid-term mini-project and a final project in an area subject to the instructor's approval.

Grading:

- Homework Assignments (8 in total equal points): 30%
- Mid-term Project (assignment for the midterm, 15-minute oral presentation): 30%
- Final Project (student will read tech. paper and implement with 15-minute oral presentation): 40%

Course Grading:

$90 \leq A \leq 100$

$80 \leq B < 90$

$70 \leq C < 80$

$60 \leq D < 70$

$F < 60$

Class Policy:

General: Attendance at the lecture is expected. If you miss a class, you are responsible for all assignments and material covered. You are required to participate in all class discussions. You will be required to answer questions or discuss your solutions in class. You must maintain good class notes and should review past materials covered before attending a class.

Homework: There will be 30% credits for 8 homework assignments. Typically, problems will be assigned in class or Emailed. These assignments are to help you determine your level of mastery of the knowledge presented in class.

Projects: There are two projects. One midterm project and one extensive final project are to be presented to the class. The projects will require performing a literature search on research topics, understanding technical papers, implementing and exploring new solutions.

Project Dates:

The dates for the two projects are shown in the class schedule. Any date changes will be finalized in class. In case you miss a class, make sure you stay in touch with important announcements.

Help session: If you attended the lectures and did not understand any material, see the instructor promptly – before the next lecture. If you did not attend the class, first obtain the notes from your classmates, review the material, and then see your instructor for further clarification.

Plagiarism: Plagiarism will be severely penalized and may result in an F grade for the course or you may receive no credit for the specific projects. Students are expected to exhibit the same level of professionalism and integrity that will distinguish them in their future careers. Both the person who reproduced in whole or in any part from the work of others and the person who allowed the work to be copied will be penalized. Consequences and procedures for dealing with cases of academic dishonesty are outlined in the JHU catalog on the Student Code of Rights and Responsibilities.

Attendance Policy:

Consistent with JHU guidelines, students absent from regularly scheduled project presentations because of authorized University activities will have the opportunity to take them at an alternate time. Make-up project presentations for absences due to any other reason will be at the discretion of the instructor.

Inclusivity Statement:

JHU community is committed to creating and fostering a positive learning and working environment based on open communication, mutual respect, and inclusion.

Integrity Statement:

The integrity of the classes offered by any academic institution solidifies the foundation of its mission and cannot be sacrificed to expediency, ignorance, or blatant fraud. Therefore, I will enforce rigorous standards of academic integrity in all aspects and assignments of this course. Should you have any questions about possibly improper research citations or references, or any other activity that may be interpreted as an attempt at academic dishonesty, please see me before the assignment is due to discuss the matter.

Disclaimer: The professor reserves the right to make changes to the syllabus. Any changes that are made will be in, what the professor deems, in the best interests of the class.

Tentative Class Schedule and Topics:

Week 1: Introduction and review of classical artificial neural network architectures (MLP, CNN)

Week 2: Review of the classical convolutional neural network-based architecture (LeNet, AlexNet, VGGNet) and the functions of its layers.

Week 3: Details of Inception Architectures for GoogLeNet,

Week 4: Details of ResNet, and DenseNet architectures

Week 5: Details of Facenet, DeepFace, HighwayNet, and FractalNet, *mini-project assignment*

Week 6: Regularization theory, sparsity, and overfitting problems

Week 7: Introduction to sparse auto-encoders, Coupled CNN, and Siamese auto-encoders

Week 8: Mid-term project with oral presentation, *Mid-term Project Due*

Week 9: Generative adversarial Networks (GAN) and its applications, *Final project assignment*

Week 10: Conditional GAN, Cycle GAN, StyleGAN, and StarGAN with their applications

Week 11: Recurrent network and LSTM networks

Week 12: Data & classifier fusion, multimodal fusion

Week 13: Use of deep learning in an image, video processing, and biometrics (i.e., video super-resolution, face recognition, fingerprints, voice, iris)

Week 14: Applications of deep learning for data-mining and image retrieval using deep hashing

Week 15: Final project with oral presentation, *Final project*